

QMC Code-Validation Platform — Summary Report

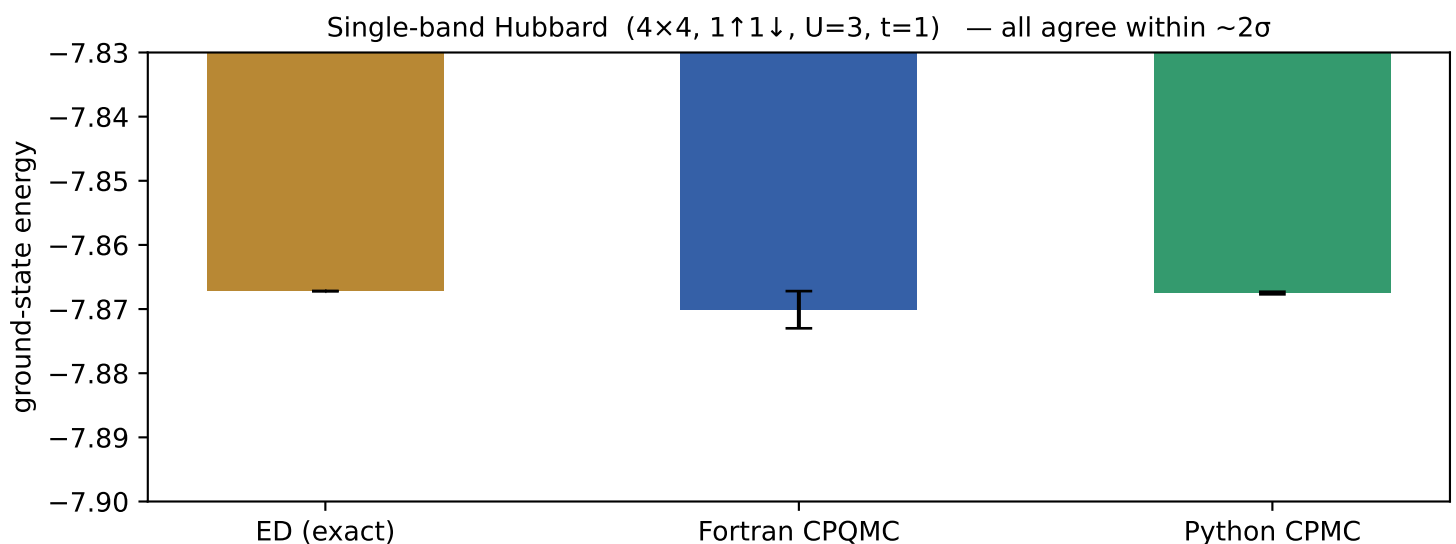
Constrained-path QMC for a two-orbital altermagnetic Hubbard model · cross-checked against exact diagonalization

Purpose

A reproducible platform to validate changes to a constrained-path quantum Monte Carlo (CPQMC) code, supporting the PRL study of enhanced d-wave pairing in a strongly correlated altermagnet. The platform compares three independent solvers on identical models: exact diagonalization (ED, ground truth on small clusters), the production Fortran CPQMC, and a from-scratch Python CPQMC port.

Platform components

- Reproducible toolchain (conda): gfortran + OpenMPI + OpenBLAS; QuSpin for ED.
- Build/run/parse harness: builds each QMC version, runs it, parses output to a common schema.
- ED reference (QuSpin): single-band Hubbard and the two-orbital altermagnet model.
- Comparison engine: per-observable delta / sigma / z-distance with PASS/FAIL verdicts.
- FastAPI + React UI: compute ED, pick QMC runs, overlay energies and the diff table.
- Python CPQMC port (pyqmc): constrained-path AFQMC, validated against ED.



Key finding (Fortran bug fixed)

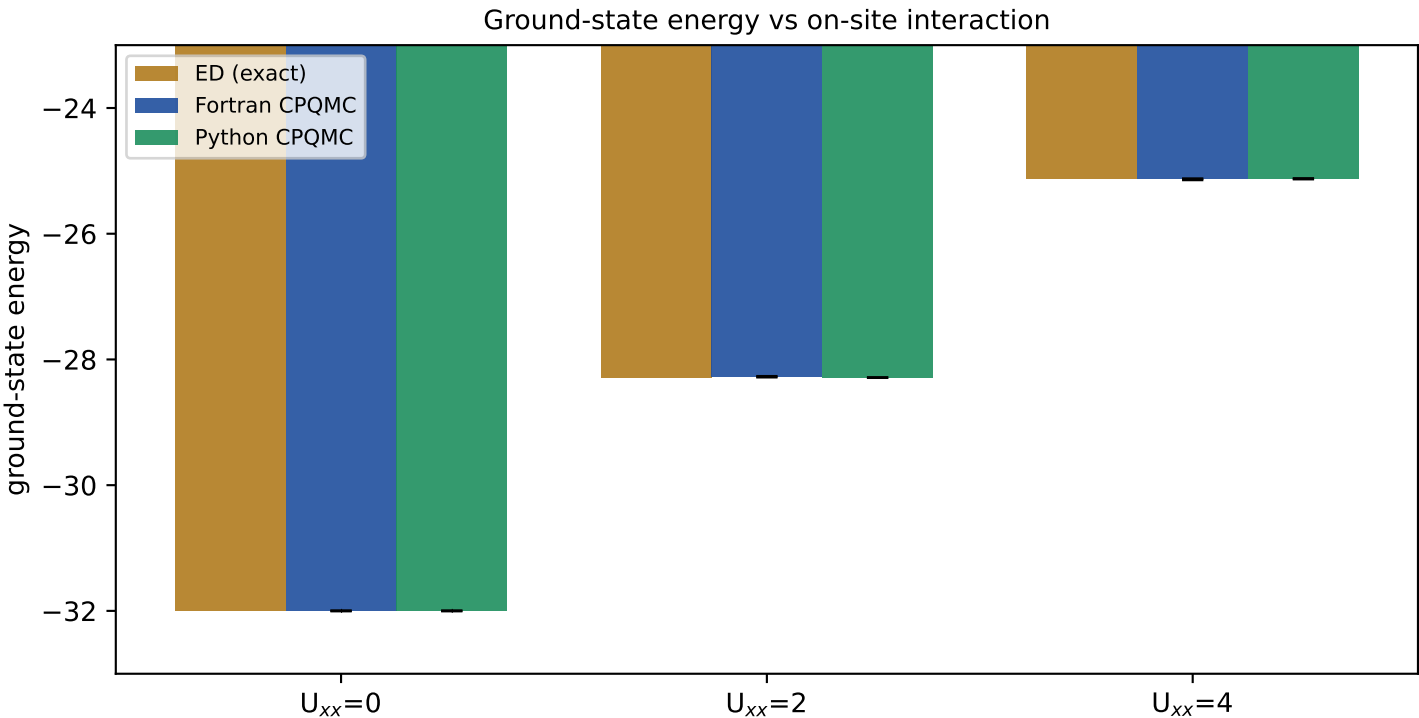
The platform localized and fixed a real sign error in the Fortran back-propagation Green's function: the measurement-phase energy came out as the exact negative of the truth (+96 vs -96 at $U=0$). Root cause: calgf method 2 built $g = +G$ while the energy estimator expects $g = \det(I-G)$. After the fix, the $U=0$ measured energy matches the exact free-fermion ground state to machine precision.

Two-Orbital Altermagnet — Three-Way Validation

Exact diagonalization vs Fortran CPQMC vs Python CPQMC on a 2x2 cluster (8 orbital-sites, 4 ↑ 4 ↓)

Model

Two-orbital (d_{xz} , d_{yz}) square lattice: $\epsilon_x = -2t_1 \cos k_x - 2t_2 \cos k_y - 4t_3 \cos k_x \cos k_y$, $\epsilon_y = (x \leftrightarrow y)$, $\epsilon_{xy} = -4t_4 \sin k_x \sin k_y$; on-site Hubbard U_{xx} . ($t_1 = t_2 = t_3 = t_4 = -1$.)



U_xx	ED (exact)	Fortran CPQMC	Python CPQMC	worst z
0	-32.000	-32.000 ± 0.000	-32.000 ± 0.000	—
2	-28.289	-28.275 ± 0.007	-28.289 ± 0.001	1.85 / 0.33
4	-25.128	-25.134 ± 0.010	-25.126 ± 0.005	0.58 / 0.42

Conclusion

ED, the production Fortran CPQMC, and the independent Python CPQMC port agree on the two-orbital altermagnet ground-state energy across interaction strengths (all within $\sim 2\sigma$) — the cross-validation benchmark requested in review. Next: validate the inter-orbital (U_{xy}) and neighbour (V) channels, then scale to publication-size clusters.